

International Cybersecurity and Digital Forensics Academy (ICDFA)

CIP-B102 Foundations of Computer and Digital Forensics

Lab 6: Data Carving with xxd, Binwalk and Scalpel

Practical forensic lab on file signatures, hex analysis, embedded-file extraction and deleted-file carving

Important pre-lab instruction: This slide is already a walkthrough. Students must first go through the full Data Carving slide material, read the explanations carefully, and understand each demonstrated command before proceeding to this lab. Do not jump straight into commands without reading the material.

Item	Details
Course Code	CIP-B102
Course Title	Foundations of Computer and Digital Forensics
Lab Number	Lab 6
Lab Theme	Data carving, file signatures, binary inspection, and recovery of deleted/embedded files
Recommended Environment	Kali Linux or Ubuntu-based forensic VM; work only on lab files or instructor-provided evidence
Submission Format	One PDF report plus recovered evidence files in a ZIP archive

1. Lab Scenario

A storage image contains files that may no longer be recoverable through normal file-system browsing. Some file metadata may be missing, damaged, or no longer reliable. As a junior digital forensic analyst, your task is to examine file content directly, identify file signatures, recover embedded image content, and carve deleted files from a USB image while documenting every step in a repeatable way.

This lab focuses on forensic analysis only. It must be performed in an authorized training environment using the provided lab files.

2. Learning Outcomes

- Explain why data carving is needed when file-system metadata is missing, damaged, or unreliable.
- Use xxd to view file content in hexadecimal and identify common file headers and footers.
- Identify JPEG header and tail signatures, including SOI 0xFFD8 and EOI 0xFFD9.
- Convert a plain hexdump back into a binary file and verify that the recovered file opens correctly.
- Use The Sleuth Kit commands such as istat, icat, blkcat and fls to examine file and sector-level content.
- Use binwalk to locate and extract embedded files from another file.
- Use scalpel to carve deleted files from a USB image and review its audit log.
- Prepare a professional forensic lab report with commands, screenshots, hashes, findings and limitations.

3. Required Downloads and Reference URLs Extracted from the Slide

URL handling: The URLs below were extracted from the uploaded Data Carving slide material. Internal PowerPoint/OpenXML schema URLs are intentionally excluded because they are not lab resources.

Resource	Exact URL	Purpose
DD image for FAT/TSK sector demonstration	https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd	Used for direct access to data units with icat and blkcat.

JPEG sample file	https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_file/J_ub_law.jpg	Used for xxd, header/tail signature analysis and hexdump reconstruction.
USB carving archive	https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_image/120M.7z	Used for deleted-file carving with scalpel.
TSK istat reference	https://wiki.sleuthkit.org/index.php?title=Istat	Reference for file metadata/statistics command.
JPG/JPEG structure reference	https://www.offshoreclippingpath.com/difference-between-jpg-and-jpeg/	Reference for JPG/JPEG file structure discussion.
JPEG Huffman coding reference	https://www.print-driver.com/stories/huffman-coding-jpeg	Reference for compression discussion in the slide.

4. Tools Required

Tool	Use in this Lab
Kali Linux/Ubuntu VM	Recommended forensic working environment.
xxd	Hexdump and reverse-hexdump tool.
The Sleuth Kit	Commands such as istat, icat, blkcat, fls and mmls.
binwalk	Detects and extracts embedded file signatures.
scalpel	File-carving tool based on configured headers and footers.
7z/p7zip	Extracts the downloaded 120M.7z archive.
imagemagick/display or any image viewer	Used to confirm recovered JPG files open correctly.
hash tools	md5sum and sha256sum for integrity verification.

Install missing tools as needed:

```
sudo apt update
sudo apt install -y sleuthkit binwalk scalpel p7zip-full xxd imagemagick
```

5. Evidence Handling and Safety Rules

1. Work only on downloaded lab files or instructor-provided evidence. Do not run carving tools on unauthorized devices.
2. Create a dedicated working directory and keep original evidence files read-only where possible.
3. Record hash values before and after extraction where applicable.
4. Do not modify the original evidence image. Any experimental modification must be done only on a copied file.
5. Every screenshot must show the command entered, command output, date/time where possible, and the student name or working folder path.

6. Preparation: Create Working Folder and Download Files

Create a clean working directory for the lab:

```
mkdir -p ~/CIP-B102_Lab6_Data_Carving
cd ~/CIP-B102_Lab6_Data_Carving
```

Download the slide-provided files:

```
wget -q https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd -O Ch01InChap01.dd
wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_file/J_ub_law.jpg -O J_ub_law.jpg
wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_image/120M.7z -O 120M.7z
```

Verify that the files downloaded successfully:

```
ls -lh
file Ch01InChap01.dd J_ub_law.jpg 120M.7z
md5sum Ch01InChap01.dd J_ub_law.jpg 120M.7z
sha256sum Ch01InChap01.dd J_ub_law.jpg 120M.7z
```

Screenshot required: Capture the download folder, file sizes, file command output and hash values. These are required in your final report.

7. Part A: Understand How File Content Is Identified

In your own words, explain the difference between retrieving a file through file-system metadata and recovering a file by carving. Your explanation must include the terms file metadata, header signature, footer/tail signature, sector/block, and unallocated space.

Answer these questions in your report:

- Why can a deleted file still be partially or fully recoverable?
- Why can a file name be lost even when file content is recovered?
- What is the difference between normal file recovery and data carving?
- What JPEG markers indicate the beginning and end of a JPEG file?

8. Part B: Examine a JPG File with xxd

Use xxd to inspect the binary content of the downloaded JPG sample.

```
xxd J_ub_law.jpg | head
xxd J_ub_law.jpg | tail
xxd -l 120 J_ub_law.jpg
```

Search for common JPG signatures in a plain hexdump:

```
xxd -p J_ub_law.jpg > J_ub_law_hexdump.txt
grep -o "ffd8" J_ub_law_hexdump.txt | head
grep -o "ffd9" J_ub_law_hexdump.txt | tail
```

Create a continuous hexdump and search across line breaks:

```
xxd -p J_ub_law.jpg | tr -d "\n" > J_ub_law_hexdump_singleline.txt
grep -o "ffd8" J_ub_law_hexdump_singleline.txt
grep -o "ffd9" J_ub_law_hexdump_singleline.txt
```

Recover the image back from the plain hexdump and compare hashes:

```
xxd -r -p J_ub_law_hexdump.txt J_ub_law_reverse_dump.jpg
file J_ub_law_reverse_dump.jpg
md5sum J_ub_law.jpg J_ub_law_reverse_dump.jpg
sha256sum J_ub_law.jpg J_ub_law_reverse_dump.jpg
```

Expected observation: If the reverse-dumped JPG is reconstructed correctly, the hash of J_ub_law.jpg and J_ub_law_reverse_dump.jpg should match.

9. Part C: Inspect Metadata-Like Strings Inside the JPG

The slide demonstrates that useful evidence can appear inside file headers and metadata regions. Inspect the JPG around the region shown in the material:

```
xxd -s 0x100 -l 0x9f J_ub_law.jpg
strings J_ub_law.jpg | head -50
strings J_ub_law.jpg | grep -Ei "nikon|photoshop|adobe|date|copyright|author|name"
```

In your report, record any visible camera/software/name/date-related strings found. If no string is found for a term, state “not found” instead of guessing.

10. Part D: Directly Access Data Units in a DD Image Using TSK

Use the Ch01InChap01.dd image to review direct access to file-system data. This reinforces the idea that forensic tools can examine content without normal file browsing.

```
img_stat Ch01InChap01.dd
mmls Ch01InChap01.dd
fsstat Ch01InChap01.dd | head -40
fls Ch01InChap01.dd
fls -d Ch01InChap01.dd
```

Use icat and blkcat examples. Replace inodeID and sectorID with values you identify from your own command output.

```
# Example syntax - replace values after you identify them
icat Ch01InChap01.dd inodeID > recovered_from_inode.bin
blkcat Ch01InChap01.dd sectorID 1 > recovered_sector.bin
file recovered_from_inode.bin recovered_sector.bin
xxd recovered_sector.bin | head
```

Do not fake values: Students must show their actual inode or sector values from command output. Marks will be deducted for copying placeholders without evidence.

11. Part E: Embedded File Discovery and Extraction with Binwalk

Use binwalk to identify embedded files or signatures. The slide shows an example where an embedded JPG can be manually carved from a Word document. The command demonstrated in the slide is shown below for instructor-provided File_carving.docx:

```
binwalk File_carving.docx
# Manual extraction example from slide; use only if File_carving.docx is provided by your
instructor
dd if=File_carving.docx skip=3934 of=cat_again.jpg count=85046 bs=1
file cat_again.jpg
```

The slide did not provide a direct download URL for File_carving.docx. Therefore, if your instructor does not provide that file, complete the binwalk practice using the sample JPG and the USB carving image/archive from this lab, and document that File_carving.docx was not available in the slide download URLs.

Automatic extraction example:

```
binwalk -e File_carving.docx
ls -lah _File_carving.docx.extracted 2>/dev/null || true
```

12. Part F: Prepare and Examine the USB Image for Carving

Extract the USB image archive and verify the extracted file.

```
7z l 120M.7z
7z e 120M.7z
ls -lh
file * | tee file_types_lab6.txt
md5sum * | tee md5_after_extract_lab6.txt
```

Identify the extracted USB image name. The slide refers to usb_fat_carving.001. If your extracted filename is different, use your actual filename in the commands below.

```
IMAGE="usb_fat_carving.001"
ls -lh "$IMAGE"
```

```
img_stat "$IMAGE"
mmls "$IMAGE"
fls -o 128 "$IMAGE"
fls -o 128 -d "$IMAGE"
```

Offset note: The slide uses offset 128. Students must still confirm partition layout with mmls and explain why that offset is used.

13. Part G: Configure Scalpel for File Carving

Scalpel uses a configuration file that defines which file types to carve. Open the configuration file and enable JPG and other instructor-approved file types.

```
sudo cp /etc/scalpel/scalpel.conf /etc/scalpel/scalpel.conf.bak
sudo nano /etc/scalpel/scalpel.conf
```

Inside the configuration file, enable only the file types required by this lab, especially JPG/JPEG. Save and close the file. Capture a screenshot showing the enabled JPG line(s).

Run scalpel on the USB image:

```
rm -rf output_lab6_scalpel
scalpel "$IMAGE" -o output_lab6_scalpel
find output_lab6_scalpel -type f -maxdepth 3 -print
cat output_lab6_scalpel/audit.txt | head -80
```

Review recovered files:

```
find output_lab6_scalpel -type f -exec file {} \;
find output_lab6_scalpel -type f -exec md5sum {} \; > carved_files_md5.txt
find output_lab6_scalpel -type f -name "*.jpg" -o -name "*.JPG"
```

Visual verification: Open at least two carved JPG images and include screenshots in your report. If no image opens, explain the error and include file command output.

14. Part H: Document Findings and Limitations

For every carved file, document what was recovered and what was not recovered. Remember that carving may recover content, but it may not recover the original filename, full path, timestamps, or complete metadata.

Finding Item	Student Response
Original evidence file name	
Original evidence hash values	
Partition offset used	
Scalpel configuration file types enabled	
Number of carved files recovered	
Number of carved JPG files that opened successfully	
Limitations observed	

15. Required Screenshots

- ☐ Slide material reviewed before starting the lab.
- ☐ Working folder created and all downloaded files listed with sizes.
- ☐ Hash values for Ch01InChap01.dd, J_ub_law.jpg and 120M.7z.
- ☐ xxd output showing JPG header and tail investigation.
- ☐ Reverse dump reconstruction and hash comparison.
- ☐ strings/metadata-style search output from J_ub_law.jpg.

- ☐ mmls/fls output for the DD or USB image.
- ☐ Scalpel configuration showing enabled file type(s).
- ☐ Scalpel run output and audit.txt.
- ☐ At least two recovered JPG files opened successfully, or error evidence if none opens.

16. Student Deliverables

- A PDF lab report named CIP-B102_Lab6_Firstname_Lastname.pdf.
- A ZIP archive containing recovered/carved files and hash text files.
- A command log or appendix containing all commands used.
- A short reflection of not more than 250 words explaining why data carving cannot always recover filenames and metadata.

17. Report Template

6. Title page: student name, registration number, course code, lab number and date.
7. Objective: explain the purpose of Lab 6.
8. Tools and environment: OS, VM, tool versions, working directory.
9. Evidence sources: exact URLs used and hash values of downloaded files.
10. Procedure: step-by-step commands and screenshots.
11. Findings: recovered files, signatures, metadata strings, audit log summary.
12. Validation: hash values and file command output.
13. Limitations: missing filenames, possible incomplete carved files, unavailable instructor-only files.
14. Conclusion: what was learned from file carving.

18. Marking Rubric (100 Marks)

Assessment Area	Marks
Pre-lab understanding and safety compliance	10
Correct download, organization and hashing of evidence files	15
Correct use of xxd, hexdump conversion and JPG signature analysis	15
Correct TSK examination and explanation of inode/sector-level access	15
Correct binwalk/scalpel carving workflow	20
Quality of screenshots, command evidence and recovered-file verification	15
Professional report structure, findings and limitations	10

19. Academic Integrity and Authorization Statement

By submitting this lab, the student confirms that all work was performed on the provided training files or instructor-approved evidence only, and that the report represents their own practical work. Any attempt to submit copied screenshots, fabricated hash values, or commands not executed by the student may result in disciplinary action.